

Companion to the appendix: full derivations

This document carries out, step by step, the derivations that the appendix quotes as results. The section numbers are the ones the appendix refers to. Nothing here is new physics; it exists so that every result in the appendix can be checked without reconstructing it.

Contents

1	Notation and conventions	1
2	The centring criterion	2
2.1	What “centred” has to mean	2
2.2	The three symmetries of the profile	2
2.3	Derivation	2
2.4	Conjugate, not mirrored — and the self-paired middle tone	3
2.5	Structure of the admissible family	3
3	The crest factor	4
3.1	The denominator does not depend on the phases	4
3.2	Bounds	4
3.3	A ramp is a time shift, and therefore crest-neutral	4
3.4	$N = 3$: the criterion forces the worst crest factor	5
3.5	$N = 4$, and the price of abandoning the criterion	6
4	Mode integrals, the overlap, and the pulse area	6
4.1	The power of one spot	6
4.2	The overlap integral	7
4.3	Where the sinc comes from	8
4.4	Atom-weighted averages	8
5	Diffraction	8
5.1	Where the $\sin^2$ and the $\sqrt{P}$ come from	8
5.2	Many tones: coupled waves and the exact reduction	9
5.3	Why the naive sum is wrong here	10
5.4	Validity	10
5.5	The crest factor in the diffraction	10

1 Notation and conventions

Tones are indexed $n = 0, \dots, N - 1$ per axis, with equidistant frequencies $f_n = f_{\text{off}} + n d$ (d = tone spacing, not to be confused with the beat order, which only appears in Sec. 4.3). The *centred* tone index

$$\hat{n} := n - \frac{N - 1}{2} \quad \implies \quad f_n - \bar{f} = d \hat{n} \quad (1)$$

is used throughout, because it makes the frequency offset and the phase share one factor. The complex envelope of the RF signal is

$$s(t) = \text{Re} \left[A(t) e^{i2\pi \bar{f} t} \right], \quad A(t) = \sum_n a_n e^{i(2\pi d \hat{n} t + \varphi_n)}, \quad (2)$$

with a_n the RF *voltage* amplitudes. The profile parameters r_x, r_y are *power* ratios, so the edge tones carry $\varrho = \sqrt{r}$ in voltage. $|A|$ has period $T_{\text{env}} = 1/d$; for even N the index $\hat{n}$ is half-integer and $A(t + T_{\text{env}}) = -A(t)$, but $|A|$ is periodic either way, and only $|A|$ enters below.

2 The centring criterion

2.1 What “centred” has to mean

The atom sits at $\mathbf{r} = 0$, smeared over $\sigma \approx 108 \text{ nm}$ [1]. The damaging situation is not a beating pattern but a pattern that sits *off to one side* during the pulse: the pulse area is then larger on one edge of the position distribution than on the other, the rotation angle depends on where the atom happened to be, and no calibration of the mean repairs it. The requirement is therefore

$$I(-\mathbf{r}, t) = I(\mathbf{r}, t) \quad \text{for all } t, \quad (3)$$

which implies $\nabla I|_0 = 0$ at all times.

2.2 The three symmetries of the profile

With u the (even) single-spot mode and $\bar{n} = N - 1 - n$ the mirror index,

$$\mathbf{c}_{\bar{n}\bar{m}} = -\mathbf{c}_{nm} \quad (\text{positions: the grid is centred}), \quad (4)$$

$$a_{\bar{n}} = a_n, \quad b_{\bar{m}} = b_m \quad (\text{amplitudes: } [\varrho, 1, \dots, 1, \varrho]), \quad (5)$$

$$f_{\bar{n}} = F - f_n, \quad g_{\bar{m}} = G - g_m \quad (\text{frequencies}). \quad (6)$$

Equation (6) is immediate from the raster: $f_n + f_{\bar{n}} = 2f_{\text{off}} + (N - 1)d =: F$, independent of n .

2.3 Derivation

Start from $E(\mathbf{r}, t) = \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{-i2\pi(f_n + g_m)t} e^{i(\varphi_n + \psi_m)}$ and evaluate it at $-\mathbf{r}$, relabelling the sum by the mirror index (a permutation, so nothing is lost).

Step 1 (positions; u even). $-\mathbf{r} - \mathbf{c}_{\bar{n}\bar{m}} = -\mathbf{r} + \mathbf{c}_{nm} = -(\mathbf{r} - \mathbf{c}_{nm})$, and u even gives $u(-\mathbf{r} - \mathbf{c}_{\bar{n}\bar{m}}) = u(\mathbf{r} - \mathbf{c}_{nm})$: the same envelope factor as in $E(\mathbf{r}, t)$.

Step 2 (amplitudes). $a_{\bar{n}} b_{\bar{m}} = a_n b_m$, unchanged.

Step 3 (frequencies) — the decisive one.

$$e^{-i2\pi(f_{\bar{n}} + g_{\bar{m}})t} = e^{-i2\pi(F + G)t} \underbrace{e^{+i2\pi(f_n + g_m)t}}_{\text{the conjugate of the original}}. \quad (7)$$

The frequency mirror does not reproduce the time factor, it *conjugates* it, up to one global factor that is the same for every spot.

Step 4 (collect). Comparing

$$E(-\mathbf{r}, t) = e^{-i2\pi(F+G)t} \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{+i2\pi(f_n+g_m)t} e^{i(\varphi_{\bar{n}}+\psi_{\bar{m}})}, \quad (8)$$

$$E^*(\mathbf{r}, t) = \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{+i2\pi(f_n+g_m)t} e^{-i(\varphi_n+\psi_m)}, \quad (9)$$

the two sums run over the same basis functions with the same real coefficients and differ only in a constant phase per term. They are proportional — and hence $I(-\mathbf{r}, t) = I(\mathbf{r}, t)$ — if and only if $e^{i(\varphi_{\bar{n}}+\psi_{\bar{m}})} = e^{i\Theta} e^{-i(\varphi_n+\psi_m)}$ with a *single* Θ . Separating the n - and m -dependence,

$$\boxed{\varphi_n + \varphi_{N-1-n} = c \quad \forall n, \quad \psi_m + \psi_{M-1-m} = c' \quad \forall m.} \quad (10)$$

2.4 Conjugate, not mirrored — and the self-paired middle tone

Equation (10) is often misread as $\varphi_{\bar{n}} = \varphi_n$. That is a different condition and it does not centre the pattern: the mirror conjugates, so the phases must *sum* to a constant. Numerically, mirror-symmetric sets leave centroid offsets of 44 nm to 84 nm against $\sigma = 108$ nm; sets obeying (10) give 0.00 nm. The two coincide only for real phasors, $\varphi_n \in \{0, \pi\}$, which is where the confusion comes from.

A natural objection is that for odd N the middle tone has no partner. It has: the map $n \mapsto \bar{n} = N - 1 - n$ is an involution, and involutions have pairs *and fixed points* ($N = 3$: $0 \leftrightarrow 2$, $1 \mapsto 1$; $N = 4$: none; $N = 5$: $2 \mapsto 2$). A fixed point is not unpaired, it is paired with itself, and (10) holds for every n , so

$$2\varphi_{(N-1)/2} = c \quad \implies \quad \varphi_{(N-1)/2} = \frac{c}{2} \quad \text{or} \quad \frac{c}{2} + \pi. \quad (11)$$

The objection implicitly assumes each pair may carry its own constant. It may not: c is the single global Θ , and one $e^{i\alpha(t)}$ has to serve all spots at once. The middle tone sits at $\bar{f}$, so its spot lies at the mirror centre and maps onto itself.

Numerical check. With $\varphi_x = (0, \psi, 0)$ — so $c = 0$ — and $\varphi_y = (0, 98, 16, 114)^\circ$ held fixed (which satisfies its own criterion, $0 + 114 = 98 + 16$), the largest intensity gradient at the atom over a full beat period is

ψ	$\max_t \partial_x I /I$	$\max_t \partial_y I /I$	
0°	$4 \cdot 10^{-10}$	$4 \cdot 10^{-10}$	criterion satisfied
10°	$8.8 \cdot 10^{-4}$	$2.6 \cdot 10^{-4}$	
30°	$2.5 \cdot 10^{-3}$	$7.4 \cdot 10^{-4}$	
90°	$5.1 \cdot 10^{-3}$	$1.5 \cdot 10^{-3}$	
180°	$4 \cdot 10^{-10}$	$4 \cdot 10^{-10}$	criterion satisfied

The gradient vanishes at exactly the two values predicted by (11) and nowhere else; with $c = 60^\circ$ the zeros move to 30° and 210° . Note the third column: violating the criterion in x also tilts the y gradient. With a separable (Gaussian) envelope it would not ($3 \cdot 10^{-10}$ in every row); the real Airy envelope is radial, hence not separable, and couples the axes. The criterion must hold on *both*.

2.5 Structure of the admissible family

Writing $\varphi_n = c/2 + \chi_n$ turns (10) into

$$\chi_{\bar{n}} = -\chi_n, \quad (12)$$

a global phase plus an antisymmetric part. Antisymmetric vectors in $\mathbb{R}^N$ span $\lfloor N/2 \rfloor$ dimensions, so the criterion halves the phase freedom, from $N - 1$ to $\lfloor N/2 \rfloor$. One of those is always a ramp and therefore crest-neutral (Sec. 3.3), leaving $\lfloor N/2 \rfloor - 1$ crest-relevant dimensions — zero for $N = 3$.

3 The crest factor

3.1 The denominator does not depend on the phases

With $C = \max_t |s| / \text{rms}_t[s]$ and the two time scales separated ($\langle s^2 \rangle = \frac{1}{2} \langle |A|^2 \rangle$, while the peak of $|s|$ is the peak of $|A|$),

$$C = \sqrt{2} \frac{\max_t |A(t)|}{\text{rms}_t |A(t)|}. \quad (13)$$

Expanding $\langle |A|^2 \rangle$ gives a double sum whose off-diagonal terms are phasor averages,

$$\langle e^{i[2\pi\Delta f t + \Delta\varphi]} \rangle_T = e^{i\Delta\varphi} \frac{e^{i2\pi\Delta f T} - 1}{i2\pi\Delta f T}, \quad (14)$$

which vanishes *exactly* when $\Delta f T$ is a non-zero integer: a phasor completing whole turns averages to the centre of the circle *regardless of where it starts*, and that is precisely why $\Delta\varphi$ drops out. With $T = T_{\text{env}} = 1/d$ every $\Delta f = f_n - f_m$ is an integer multiple of d , so

$$\boxed{\text{rms}_t |A| = \sqrt{\sum_n a_n^2} \quad \text{for every phase set.}} \quad (15)$$

If the window is chosen shorter, (14) no longer vanishes and the rms becomes phase dependent: for the x axis, using $T = 1/\Delta f$ of the neighbour pair instead of $1/d$, the rms varies between 1.548 and 2.111 for different phase sets, against 1.714643 always with the correct window.

3.2 Bounds

$|A(t)| \leq \sum_n a_n$ at all times (triangle inequality), and $|A|$ reaches at least its own rms somewhere, so

$$\sqrt{2} \leq C \leq \sqrt{2} \frac{\sum_n a_n}{\sqrt{\sum_n a_n^2}} \xrightarrow{a_n \equiv 1} \sqrt{2N}. \quad (16)$$

The upper bound is attained exactly when all phasors align at some instant. Which sets approach the lower bound is classical: Schroeder's quadratic phases reach $C \approx 1.7$ for large N [2], Kitayoshi's construction is comparable [3], and the attainable minimum is studied systematically in [4]. *None of them satisfies Eq. (10)*, which is the whole tension.

3.3 A ramp is a time shift, and therefore crest-neutral

A ramp is $\varphi_n = \varphi_s + \delta n$. The start value is common to all tones and factors out; subtracting the mean $\bar{\varphi} = \varphi_s + \delta(N-1)/2$ leaves

$$\chi_n := \varphi_n - \bar{\varphi} = \delta \hat{n}. \quad (17)$$

This is not a replacement but a factorisation: with $\varphi_n = \bar{\varphi} + \chi_n$,

$$A(t) = e^{i\bar{\varphi}} \underbrace{\sum_n a_n e^{i(2\pi d \hat{n} t + \chi_n)}}_{=: \tilde{A}(t)}, \quad |A| = |\tilde{A}|, \quad C(A) = C(\tilde{A}). \quad (18)$$

Since $\hat{\hat{n}} = -\hat{n}$, Eq. (17) satisfies (12): **every ramp obeys the centring criterion automatically.**

Does a shift exist? The question is not how to define τ but whether one number τ — the same for all times and all tones — can satisfy $\tilde{A}_\delta(t) = \tilde{A}_0(t + \tau)$ for all t . Both sides expand in the same linearly independent functions $e^{i2\pi d\hat{n}t}$, so coefficients may be compared:

$$a_n e^{i\delta\hat{n}} = a_n e^{i2\pi d\hat{n}\tau} \iff 2\pi d\hat{n}\tau \equiv \delta\hat{n} \pmod{2\pi} \quad \text{for every } n. \quad (19)$$

This is a *system*: N conditions, one unknown. Generically unsolvable — but here the same factor $\hat{n}$ stands on both sides, cancels for $\hat{n} \neq 0$ and gives $0 = 0$ for $\hat{n} = 0$, so all N conditions collapse into one:

$$\boxed{\tau = \frac{\delta}{2\pi d} \implies \tilde{A}_\delta(t) = \tilde{A}_0(t + \tau).} \quad (20)$$

The content is the solvability, not the value of τ .

Converse. For an arbitrary mean-free set χ_n the same comparison demands $\tau = \chi_n/(2\pi d\hat{n})$, a number that generally depends on n . A common τ exists iff $\chi_n/\hat{n}$ is the same for all tones, i.e. iff $\chi_n \equiv \delta\hat{n}$ modulo 2π . **A phase set is a time shift of the envelope if and only if it is a ramp.**

Why the crest factor cannot notice. Both $\max_t$ and rms_t run over a *full* period, and $t' = t + \tau$ maps a full period onto a full period; the *set* of values of $|\tilde{A}|$ is unchanged, only its ordering. Hence $C(\delta) = C(0)$ exactly. Numerically, over 400 000 samples and $\delta = 0^\circ \dots 359^\circ$: $\max |\tilde{A}| = 2.96977156$, $\text{rms} = 1.71464282$, $C = 2.44942630$ for $N = 3$, and $4.15406592/2.07846097/2.82648385$ for $N = 4$ — agreement to eleven digits.

What a ramp does change. The shift is real against an *absolute* time origin, and the trigger t_0 is one. At fixed $t_0 = 0$ the pulse area over $1\mu\text{s}$ drops to 0.825 at $\delta = 12.5^\circ$ and to 0.043 at 90° of its value at $\delta = 0$. “Redundant with t_0 ” means it must be considered *together with* the trigger, not that it is harmless. Across two axes, only *matched* ramps ($\delta_x/d_x = \delta_y/d_y$, i.e. $\tau_x = \tau_y$) shift the whole two-dimensional pattern: checked over a full beat period and the entire grid, matched ramps give $\max |I_{\text{ramp}}(\mathbf{r}, t) - I_0(\mathbf{r}, t - \tau)| / \max I = 8 \cdot 10^{-14}$, a ramp on x alone gives 0.28.

3.4 $N = 3$: the criterion forces the worst crest factor

With $\theta := 2\pi dt$ and voltage amplitudes $a_0 = a_2 = \varrho$, $a_1 = 1$,

$$A(t) = \varrho e^{i(\varphi_0 - \theta)} + e^{i\varphi_1} + \varrho e^{i(\varphi_2 + \theta)}. \quad (21)$$

The criterion requires $\varphi_0 + \varphi_2 = c$ and $2\varphi_1 = c$, so the outer phases lie symmetrically about $c/2$: $\varphi_0 = \frac{c}{2} - \chi$, $\varphi_2 = \frac{c}{2} + \chi$, $\varphi_1 = \frac{c}{2}$ or $\frac{c}{2} + \pi$, with χ the only remaining parameter. Factoring out $e^{ic/2}$,

$$\boxed{A(t)e^{-ic/2} = \varrho e^{-i(\chi + \theta)} \pm 1 + \varrho e^{+i(\chi + \theta)} = 2\varrho \cos(\theta + \chi) \pm 1.} \quad (22)$$

The whole time dependence now sits in a *real* cosine, which sweeps $[-1, 1]$ over one period no matter what χ is — χ only sets *when*. Maximising over $u := \cos(\theta + \chi) \in [-1, 1]$,

$$\text{branch } + : \quad \max_u |2\varrho u + 1| = 2\varrho + 1 \quad \text{at } u = +1, \quad (23)$$

$$\text{branch } - : \quad \max_u |2\varrho u - 1| = |-2\varrho - 1| = 2\varrho + 1 \quad \text{at } u = -1, \quad (24)$$

so $\max_t |A| = 2\varrho + 1 = \sum_n a_n$ for both branches and every χ, c — which is the triangle-inequality bound (16), valid for *any* phase set. The admissible family therefore always attains it.

With three tones the centring criterion fixes the crest factor at the largest value the amplitudes allow, for every c , every χ and both branches. The reason is visible in (22): the only free parameter appears *inside* the cosine. It can move *when* the phasors align, not *whether* they do.

With $\varrho = \sqrt{r_x} = \sqrt{0.97} = 0.9848858$,

$$\sum_n a_n = 2\varrho + 1 = 2.9697716, \quad \sqrt{\sum_n a_n^2} = \sqrt{2r_x + 1} = \sqrt{2.94} = 1.7146428, \quad (25)$$

— note that the square sum contains the *power* ratio linearly, the square root cancelling — hence $C_x = \sqrt{2} \cdot 2.9697716 / 1.7146428 = 2.4494263$. For exactly equal amplitudes this would be $\sqrt{2N} = \sqrt{6} = 2.4494897$; the working point lies $6 \cdot 10^{-5}$ below, so the two agree to four digits without being the same number.

The family is the ramp family. The two branches $[0, c/2, c]$ and $[0, c/2 + \pi, c]$ look distinct but are the two halves of one circle: the ramp $\varphi_n = \delta n$ gives $c = 2\delta$ and $\varphi_1 = \delta$, and as δ runs once through $[0, 2\pi)$, $c = 2\delta \bmod 2\pi$ runs through it *twice* — for $\delta < \pi$ with $\varphi_1 = c/2$, for $\delta \geq \pi$ with $\varphi_1 = c/2 + \pi$. Checked at $\delta = 0^\circ \dots 315^\circ$ in 45° steps: $C_x = 2.4494$ in all eight cases.

3.5 $N = 4$, and the price of abandoning the criterion

For four tones (10) leaves $\varphi_y = [0, b, c - b, c]$ with two parameters; the ramp is $b = c/3$, and any other b genuinely reshapes the envelope. At $c = 114^\circ$:

b	φ_y [deg]	$\max A $	C_y
38°	$[0, 38, 76, 114]$	4.154	2.8265 (ramp, $= \sum a_n$)
60°	$[0, 60, 54, 114]$	4.022	2.7364
98°	$[0, 98, 16, 114]$	3.228	2.1963
114°	$[0, 114, 0, 114]$	3.639	2.4761

For completeness, what abandoning the criterion on the x axis would buy. Parametrising the deviation by $\psi = \varphi_1 - (\varphi_0 + \varphi_2)/2$ — the curvature of the phase profile, zero exactly when the criterion holds — Eq. (22) generalises to $\max_t |A| = \sqrt{4\varrho^2 + 4\varrho|\cos \psi| + 1}$, minimal at $\psi = \pm 90^\circ$. Re-optimising φ_y and t_0 at each ψ :

ψ	C_x	$(C_x C_y)^2$	rel. P_{avg}	$U(\theta)$
0° (centred)	2.4494	47.9	1.00	0.94 %
45°	2.2836	40.7	1.18	1.27 %
90°	1.8220	25.9	1.85	3.88 %

Interestingly the *centroid* stays at zero in all three rows: t_0 and the y phases can still null the odd moment of the *time-integrated* pulse area. What is lost is not the position but the quiescence — at $\psi = 0$ the pattern stands still over the whole pulse, at $\psi \neq 0$ it breathes and only the balance over the pulse cancels. The surviving *even* part is the spread $U(\theta)$, which grows by a factor four. For a π pulse, $1 - F \approx (\pi U/2)^2$, i.e. $2.2 \cdot 10^{-4}$ versus $3.7 \cdot 10^{-3}$.

4 Mode integrals, the overlap, and the pulse area

4.1 The power of one spot

With $g_s = A_s u(\mathbf{r} - \mathbf{c}_s)$, a single spot carries $\int g_s^2 d^2 r = a_s P_1$, $P_1 = \int u^2 d^2 r$. For the Gaussian, substituting $x = 2r^2/w_0^2$,

$$P_1^G = \int_0^\infty e^{-2r^2/w_0^2} 2\pi r dr = \frac{\pi w_0^2}{2} \int_0^\infty e^{-x} dx = \frac{\pi w_0^2}{2}. \quad (26)$$

For the Airy mode, with $x = kr$ and $\int_0^\infty J_\nu^2(x) dx/x = 1/(2\nu)$ [5],

$$P_1^A = \int_0^\infty \left(\frac{2J_1(kr)}{kr} \right)^2 2\pi r dr = \frac{8\pi}{k^2} \cdot \frac{1}{2} = \frac{4\pi}{k^2}, \quad k = \frac{3.8317}{1.4830 w_0}. \quad (27)$$

The Airy spot carries 1.20 times the power of a Gaussian of the same $1/e^2$ width, because of its rings.

4.2 The overlap integral

Only degenerate pairs survive the time average, and each contributes $2A_s A_{s'} \cos(\phi_s - \phi_{s'}) O(\varrho)$ with $O(\varrho) = \int u(\mathbf{r} - \mathbf{c}_s) u(\mathbf{r} - \mathbf{c}_{s'}) d^2 r$ and $\varrho = |\mathbf{c}_s - \mathbf{c}_{s'}|$.

Gaussian. Completing the square with the midpoint $\mathbf{m} = \frac{1}{2}(\mathbf{c}_s + \mathbf{c}_{s'})$,

$$e^{-\frac{(\mathbf{r}-\mathbf{c}_s)^2}{w_0^2}} e^{-\frac{(\mathbf{r}-\mathbf{c}_{s'})^2}{w_0^2}} = e^{-\frac{\varrho^2}{2w_0^2}} e^{-\frac{2(\mathbf{r}-\mathbf{m})^2}{w_0^2}} \implies O^G = P_1^G e^{-\varrho^2/2w_0^2}. \quad (28)$$

Airy: three ingredients. (i) *The mode is band limited.* The focal field is the Fourier transform of the pupil field, and a uniformly illuminated circular pupil is a disc [6, Ch. 4, 6], [7, Sec. 8.5]. Hence

$$\tilde{u}(q) = \int u(|\mathbf{r}|) e^{-i\mathbf{q}\cdot\mathbf{r}} d^2 r = \frac{4\pi}{k^2} \Theta(k - q), \quad (29)$$

a *flat* spectrum inside the cutoff. The prefactor follows at $\mathbf{q} = 0$: $\tilde{u}(0) = 2\pi \int_0^\infty u(r) r dr = \frac{4\pi}{k} \int_0^\infty J_1(kr) dr = \frac{4\pi}{k^2}$, using $\int_0^\infty J_1(x) dx = 1$ [5].

(ii) *The integral over a disc.* The angular integral gives $\int_0^{2\pi} e^{ix \cos \varphi} d\varphi = 2\pi J_0(x)$ and the radial one closes with $\int_0^X J_0(t) t dt = X J_1(X)$ (one line from $\frac{d}{dz}[z J_1(z)] = z J_0(z)$):

$$\int_{|\mathbf{q}| < k} e^{i\mathbf{q}\cdot\mathbf{d}} d^2 q = 2\pi \int_0^k J_0(q\varrho) q dq = \frac{2\pi k J_1(k\varrho)}{\varrho}. \quad (30)$$

(iii) *The correlation theorem, written out.* With the convention $\tilde{u}(\mathbf{q}) = \int u e^{-i\mathbf{q}\cdot\mathbf{r}} d^2 r$ and the whole $1/(2\pi)^2$ in the back transform [6, Ch. 2], expand *only* the shifted factor, so that the shift becomes a phase:

$$O(\mathbf{d}) = \int d^2 r u(\mathbf{r}) \frac{1}{(2\pi)^2} \int d^2 q \tilde{u}(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{r}} e^{-i\mathbf{q}\cdot\mathbf{d}} \quad (31)$$

$$= \frac{1}{(2\pi)^2} \int d^2 q \tilde{u}(\mathbf{q}) e^{-i\mathbf{q}\cdot\mathbf{d}} \underbrace{\int d^2 r u(\mathbf{r}) e^{+i\mathbf{q}\cdot\mathbf{r}}}_{=\tilde{u}(-\mathbf{q})}. \quad (32)$$

Interchanging the integrations is legitimate (u bounded and square integrable, $\tilde{u}$ of compact support, so the double integral converges absolutely). Now the symmetry: u real gives $\tilde{u}(-\mathbf{q}) = \tilde{u}^*(\mathbf{q})$, u even gives $\tilde{u}(-\mathbf{q}) = \tilde{u}(\mathbf{q})$; both together give $\tilde{u}(\mathbf{q})\tilde{u}(-\mathbf{q}) = |\tilde{u}(\mathbf{q})|^2$ and

$$O(\mathbf{d}) = \frac{1}{(2\pi)^2} \int |\tilde{u}(\mathbf{q})|^2 e^{-i\mathbf{q}\cdot\mathbf{d}} d^2 q. \quad (33)$$

The sign in the exponent is immaterial because $|\tilde{u}|^2$ is even. At $\mathbf{d} = 0$, Eq. (33) is Parseval's theorem and returns P_1 — the cheapest check of the prefactor. (“Correlation equals convolution” is the statement that $v(\mathbf{d} - \mathbf{r}) = v(\mathbf{r} - \mathbf{d})$ for even v .)

Putting them together. An indicator function is idempotent, $\Theta^2 = \Theta$, so squaring the spectrum changes nothing:

$$O^A(\varrho) = \frac{1}{(2\pi)^2} \left(\frac{4\pi}{k^2} \right)^2 \frac{2\pi k J_1(k\varrho)}{\varrho} = \frac{8\pi J_1(k\varrho)}{k^3 \varrho} = \underbrace{\frac{4\pi}{k^2}}_{P_1^A} \underbrace{\frac{2J_1(k\varrho)}{k\varrho}}_{u(\varrho)}. \quad (34)$$

The autocorrelation of the Airy mode is the mode itself, including its sign, which is negative between the first and the second ring. No such statement holds for the Gaussian, whose squared spectrum is narrower by $\sqrt{2}$ — which is exactly why O^G carries $2w_0^2$ in the exponent. Read in the pupil: the two spots are plane waves tilted against each other, i.e. a linear phase across the pupil, and averaging it over the disc is Eq. (30); the further apart, the more fringes fit and the more they cancel. Numerically (2048² grid, 40 μm): $O(0) = P_1$ exactly, $P_1^A = 4\pi/k^2$ to 1.2%, and $O(\varrho) = P_1^A u(\varrho)$ to 1.1% of the peak over $\varrho = 0.2 \dots 6 \mu\text{m}$, the residual being the grid truncating the outer rings. At $\varrho = 3.305 \mu\text{m}$, $u = +0.06308$.

4.3 Where the sinc comes from

Each Fourier order of $\tilde{I} = \sum_d D_d e^{i2\pi df_0 t}$ contributes

$$\int_{t_0}^{t_0+T_p} e^{i2\pi df_0 t} dt = \frac{e^{i2\pi df_0(t_0+T_p)} - e^{i2\pi df_0 t_0}}{i2\pi df_0}. \quad (35)$$

Pulling out the phase of the window *centre* makes the bracket symmetric, and $e^{ix} - e^{-ix} = 2i \sin x$ gives

$$= e^{i2\pi df_0(t_0+T_p/2)} \frac{\sin(\pi df_0 T_p)}{\pi df_0} = T_p e^{i2\pi df_0(t_0+T_p/2)} \text{sinc}(df_0 T_p). \quad (36)$$

The three factors read directly: T_p is the window length, the exponential says only the centre of the window matters for the phase, and the sinc is the Fourier transform of the window itself — integrating over a box is a convolution with a box, and convolution in time is multiplication in frequency. The $d = 0$ term survives always, $\text{sinc}(0) = 1$.

4.4 Atom-weighted averages

$\theta(\mathbf{r})$ is a field; the atom sees it through $W(\mathbf{r}) = (2\pi\sigma^2)^{-1} \exp[-(\mathbf{r} - \mathbf{r}_0)^2/2\sigma^2]$ with $\sigma^2 = \frac{\hbar}{2m\omega_r} \coth \frac{\hbar\omega_r}{2k_B T}$ [1]. On a uniform grid the integral is the weighted sum $\sum_i w_i \theta_i$, $w_i = W_i / \sum_i W_i$, since the cell area cancels; the code uses a dedicated fine grid of $\pm 4\sigma$, the global grid being far too coarse for $\sigma \approx 0.1 \mu\text{m}$. Because θ is *linear* in the D_d , the spatial average commutes with the sum: with $c_d = \int W D_d d^2r$ computed once, a full trigger scan is one matrix–vector product. The spread needs the map itself, $\sigma_\theta^2 = \int W \theta^2 d^2r - \langle \theta \rangle_W^2$, and the excitation must be averaged *after* the non-linear step, $\langle \sin^2(\theta/2) \rangle_W \neq \sin^2(\langle \theta \rangle_W/2)$.

5 Diffraction

5.1 Where the $\sin^2$ and the $\sqrt{P}$ come from

The RF drives a piezo transducer; the strain amplitude obeys $P_{ac} \propto S^2$, and the photoelastic effect converts strain into index modulation [8, Ch. 20],

$$\Delta n = -\frac{1}{2} n_0^3 p S \propto \sqrt{P_{ac}}. \quad (37)$$

This is the origin of the square root — it is in the conversion from power to amplitude, not in the diffraction. In the Bragg regime two waves exchange amplitude periodically, giving

$\eta = \sin^2 \nu$ with $\nu = \pi \Delta n L / (\lambda \cos \theta_B)$, hence

$$\eta(P) = \alpha \sin^2(\kappa \sqrt{P}) = \alpha \sin^2\left(\frac{\pi}{2} \sqrt{\frac{P}{P_s}}\right), \quad \kappa = \frac{\pi}{\lambda} \sqrt{\frac{M_2 L}{2H}} \sqrt{\zeta}, \quad (38)$$

which is the standard result [8, 9] and is quoted in exactly this form by the manufacturer [10]. Neither M_2 nor L/H nor the conversion ζ is known individually to useful accuracy, which is why α and P_s are measured [11].

5.2 Many tones: coupled waves and the exact reduction

Write the field as one undiffracted wave plus N diffracted ones, $E = A_u e^{i(\mathbf{k}_u \mathbf{r} - \omega_0 t)} + \sum_n A_n e^{i(\mathbf{k}_n \mathbf{r} - \omega_n t)}$, with $\omega_n = \omega_0 + \Omega_n$. A_u carries no number — it belongs to no tone; A_n is the beam diffracted by tone n , and every $\sum_n$ below runs over tones and never includes A_u . Substituting into the Helmholtz equation with $n^2 \approx n_0^2 + 2n_0 \sum_n \Delta n_n \cos(\mathbf{K}_n \mathbf{r} - \Omega_n t + \varphi_n)$ and collecting terms at ω_n gives [12, 13]

$$\frac{dA_n}{dz} = -i\kappa_n e^{i\varphi_n} A_u e^{i\Delta k_n z}, \quad \frac{dA_u}{dz} = -i \sum_n \kappa_n e^{-i\varphi_n} A_n e^{-i\Delta k_n z}, \quad (39)$$

with $\kappa_n = \kappa \sqrt{P_n}$ and $\Delta k_n = (\mathbf{k}_u + \mathbf{K}_n - \mathbf{k}_n) \cdot \hat{\mathbf{z}}$. Three readings:

1. **The tone phases drop out.** $\tilde{A}_n = A_n e^{-i\varphi_n}$ removes φ_n from both equations, and $|A_n| = |\tilde{A}_n|$: the diffraction efficiency is *exactly* phase independent, and φ_n reappears only as the phase of the diffracted wave, i.e. in the interference pattern. This is why the crest factor — a property of the *sum*, which only the broadband amplifier sees — cannot appear in the diffraction at all.
2. **Phase matching is assumed here, not later.** Setting $\Delta k_n = 0$ leaves the working equations; z is measured in units of the interaction length, so the exit is $z = 1$ and κ_n is the accumulated coupling.
3. **Unequal Δk_n break the reduction below**, because carrying $e^{-i\Delta k_n z}$ through the derivative creates a new combination with weights $\kappa_n \Delta k_n$, and the hierarchy closes only if all Δk_n are equal (then one gets the detuned two-wave solution) or zero.

Bright mode. With $\kappa_{\text{eff}} = \sqrt{\sum_n \kappa_n^2}$ and $B = \kappa_{\text{eff}}^{-1} \sum_n \kappa_n A_n$ one gets $dA_u/dz = -i\kappa_{\text{eff}} B$ and $dB/dz = -i\kappa_{\text{eff}} A_u$: a closed two-wave system.

Dark modes stay dark. Any $C = \sum_n c_n A_n$ with $\sum_n c_n \kappa_n = 0$ obeys $dC/dz = 0$ and $C(0) = 0$, so $C \equiv 0$. Of the N diffracted degrees of freedom, $N - 1$ are never excited — *this* is what makes the reduction exact rather than approximate.

Solve and return. $A_u = \cos(\kappa_{\text{eff}} z)$, $B = -i \sin(\kappa_{\text{eff}} z)$. Integrating $dA_n/dz = -i\kappa_n A_u$ with $A_n(0) = 0$, the *same* integral appears for every n :

$$A_n(z) = -i\kappa_n \int_0^z A_u(z') dz' = -i \frac{\kappa_n}{\kappa_{\text{eff}}} \sin(\kappa_{\text{eff}} z), \quad (40)$$

so the orders differ only by the prefactor. Summing at $z = 1$, $\eta_{\text{tot}} = \sum_n |A_n|^2 = \sin^2 \kappa_{\text{eff}} = 1 - |A_u|^2$, and — the step where the sum moves under the root —

$$\kappa_{\text{eff}} = \sqrt{\sum_n \kappa_n^2} = \sqrt{\sum_n \kappa^2 P_n} = \kappa \sqrt{P_{\text{tot}}}. \quad (41)$$

The squares of the couplings *are* the powers, so their quadratic sum is the ordinary sum of powers under one root. Hence

$$\boxed{\eta_{\text{tot}} = \alpha \sin^2(\kappa \sqrt{P_{\text{tot}}}), \quad \eta_n = \frac{P_n}{P_{\text{tot}}} \eta_{\text{tot}}.} \quad (42)$$

Energy is conserved exactly, and for $N = 1$ this reduces to (38). The same result, derived independently and verified experimentally on the same class of device, is reported in [11]; a treatment with optimised tone phases is [14].

5.3 Why the naive sum is wrong here

$\sum_n \sin^2(\kappa \sqrt{P_n})$ is not wrong physics but the small-signal limit taken in the wrong place: for $\kappa \sqrt{P} \ll 1$ both expressions tend to $\kappa^2 P_{\text{tot}}$. They separate exactly where we work:

P_{tot}	$\kappa \sqrt{P_{\text{tot}}}$	η_{tot} exact	naive sum (3 tones)
5 mW	0.066	0.00432	0.00433 (+0.10 %)
50 mW	0.208	0.0427	0.0431 (+0.97 %)
0.664 W	0.758	0.473	0.539 (+14.0 %)

5.4 Validity

Equation (42) is exact at any coupling strength (checked against direct integration to 10^{-13} , including beyond the first maximum where back-conversion sets in). What is assumed is:

1. **Common phase matching.** For $\Delta k \neq 0$ the two-wave solution becomes $\eta(\xi) = \nu^2(\nu^2 + \xi^2)^{-1} \sin^2 \sqrt{\nu^2 + \xi^2}$ with $\nu = \kappa \sqrt{P}$, $\xi = \Delta k L / 2$ [12]. Calibrating ξ against the specified 36 MHz bandwidth (half efficiency at the band edge, $\xi_{1/2} = 1.3625$ at $\nu = 0.758$) gives $\xi = 1.3625 \delta f / 18 \text{ MHz}$ and hence 0.007 % loss for the 0.37 MHz span of the working point, 1.8 % for 6 MHz, and 15 % for 18 MHz.
2. **Bragg regime, two waves per grating;** higher orders are strongly mismatched.
3. **Linear transducer.** The instantaneous strain carries the crest factor as well, so a high C also brings on acoustic nonlinearity — a second channel, independent of the amplifier, pointing the same way, not quantified here.
4. **Isotropic diffraction.** The device diffracts anisotropically in a birefringent crystal; the isotropic result agrees qualitatively [15].
5. **Independent axes.** Measurement shows a cross term, $\eta_{ij} = \eta_i^{(x)} \eta_j^{(y)} \Xi_{ij}$ with $\Xi = (1 - \xi_0(\nu_i^{(x)} - \xi_x)^2(\nu_j^{(y)} - \xi_y)^2)^2$, falling to 0.80 in the corners of the scan range [11]. Over the 0.37 MHz span used here Ξ is common to all spots.

5.5 The crest factor in the diffraction

Combining the back-off rule with (42),

$$\eta_{\text{axis}}(C) = \alpha \sin^2 \left(\frac{\pi}{2C} \sqrt{\frac{P_{\text{1dB}}}{P_s}} \right). \quad (43)$$

With the test-sheet parameters ($\alpha = 1$, $P_s = 2.85 \text{ W}$, $P_{\text{1dB}} = 3.98 \text{ W}$) this is $\sin^2(1.8559/C)$:

	C	P_{avg}	η_{axis}
theoretical minimum	$\sqrt{2} = 1.414$	1.99 W	0.935
x with $\psi = 90^\circ$ (criterion abandoned)	1.822	1.20 W	0.725
y , $b = 98^\circ$	2.196	0.83 W	0.559
$\mathbf{x}$, centred, $\mathbf{N} = 3$	2.449	0.66 W	0.472
y , ramp	2.827	0.50 W	0.373

Had both axes reached $C = \sqrt{2}$, the profile efficiency would be 87 % instead of 26.4 % — a factor 3.3. That factor is the price of the crest factor, and Sec. 3.4 says how much of it is negotiable: on the x axis none, as long as three tones are used and the pattern must stay centred.

References

- [1] C. Tuchendler, A. M. Lance, A. Browaeys, Y. R. P. Sortais, and P. Grangier. Energy distribution and cooling of a single atom in an optical tweezer. *Physical Review A*, 78(3): 033425, 2008. doi: 10.1103/PhysRevA.78.033425.
- [2] M. R. Schroeder. Synthesis of low-peak-factor signals and binary sequences with low autocorrelation. *IEEE Transactions on Information Theory*, 16(1):85–89, 1970. doi: 10.1109/TIT.1970.1054411.
- [3] H. Kitayoshi, S. Sumiea, K. Siirakawa, and S. Takeshita. DSP synthesized signal source for analog testing stimulus and new test method. In *Digest of Papers, International Test Conference*, pages 825–834. IEEE, 1985.
- [4] Stephen Boyd. Multitone signals with low crest factor. *IEEE Transactions on Circuits and Systems*, 33(10):1018–1022, 1986. doi: 10.1109/TCS.1986.1085837.
- [5] NIST. NIST digital library of mathematical functions, sec. 10.22. <https://dlmf.nist.gov/10.22>. accessed 17 September 2026.
- [6] Joseph W. Goodman. *Introduction to Fourier Optics*. W. H. Freeman, New York, 4 edition, 2017. ISBN 978-1-319-11916-4.
- [7] Max Born and Emil Wolf. *Principles of Optics*. Cambridge University Press, 7 (expanded) edition, 1999. ISBN 978-0-521-64222-4. Sec. 8.5.
- [8] Bahaa E. A. Saleh and Malvin Carl Teich. *Fundamentals of Photonics*. Wiley, Hoboken, NJ, 3 edition, 2019. ISBN 978-1-119-50687-4. Ch. 20, Acousto-Optics.
- [9] A. Korpel. Acousto-optics — a review of fundamentals. *Proceedings of the IEEE*, 69(1): 48–53, 1981. doi: 10.1109/PROC.1981.11919.
- [10] AA Opto-Electronic. Acousto-optics — application notes. <https://aaoptoelectronic.com/wp-content/uploads/2024/04/AA-THEORY-NOTES.pdf>. accessed 17 September 2026.
- [11] Marcel Mittenbühler. Reactive real-time synthesis of multiple tweezers for parallelized atom transport. Master’s thesis, Technische Universität Darmstadt, Institut für Angewandte Physik, 2024.
- [12] H. Kogelnik. Coupled wave theory for thick hologram gratings. *Bell System Technical Journal*, 48(9):2909–2947, 1969. doi: 10.1002/j.1538-7305.1969.tb01198.x.
- [13] Vincent Laude. General solution of the coupled-wave equations of acousto-optics. *Journal of the Optical Society of America A*, 20(12):2307, 2003. doi: 10.1364/JOSAA.20.002307.

- [14] S. N. Antonov and Yu. G. Rezvov. Efficient multiple-beam Bragg acoustooptic diffraction with phase optimization of a multifrequency acoustic wave. *Technical Physics*, 52(8):1053–1060, 2007. doi: 10.1134/S1063784207080154.
- [15] I. C. Chang. Acousto-optic devices and applications. In *Handbook of Optics*, volume 2. McGraw-Hill, 1995.